9 Polynomials And Equations

Multiplications

We have seen different easy ways to calculate the product of two numbers. Can you do this multiplication in head?

$$38 \times 32$$

How about writing 38 as 30 + 8 and 32 as 30 + 2?

Remember what we said about the product of two sums in the lesson **Multiplication Identities** in Class 9?

> To multiply a sum by a sum, each number in one sum is to be multiplied by each number in the other sum, and the products added.

We also wrote this in shorthand as an algebraic equation:

> For any four numbers x, y, u, v
>
> $$(x + y)(u + v) = xu + xv + yu + yv$$

Using this,

$$\begin{aligned} 38 \times 32 &= (30 + 8) \times (30 + 2) \\ &= (30 \times 30) + (30 \times 2) + (8 \times 30) + (8 \times 2) \end{aligned}$$

In this, the sum $(30 \times 2) + (8 \times 30)$ of the middle two products can be combined like this:

$$\begin{aligned} (30 \times 2) + (8 \times 30) &= (2 \times 30) + (8 \times 30) \\ &= (2 + 8) \times 30 \\ &= 10 \times 30 \\ &= 300 \end{aligned}$$

Thus we get

$$38 \times 32 = (30 \times 30) + (30 \times 2) + (8 \times 30) + (8 \times 2)$$
$$= 900 + 300 + 16$$
$$= 1216$$

Can't we do 48×42 like this?

$$48 \times 42 = 40^2 + ((8 + 2) \times 40) + (8 \times 2)$$
$$= 1600 + 400 + 16$$
$$= 2016$$

How about 78×72?

$$78 \times 72 = 4900 + 700 + 16 = 5616$$

We can write the multiplications done in all three examples above as a general principle:

For any number x

$$(x + 2)(x + 8) = x^2 + (8 + 2)x + 16$$
$$= x^2 + 10x + 16$$

What if we take some other numbers instead of 2 and 8 in this?

Do the multiplications below in head. Write the general principle got from each set as an algebraic identity:

(1) (i) 43×47 (ii) 63×67 (iii) 103×107

(2) (i) 51×52 (ii) 81×82 (iii) 301×302

Polynomial multiplication

What general principle do we get from the second set of problems above?

For any natural number x

$$(x + 1)(x + 2) = x^2 + 3x + 2$$

According to what we noted in the beginning about the product of sums, this is true not only for a natural number x, but for any number whatsoever. That is,

$$(x + 1)(x + 2) = x^2 + 3x + 2 \text{ for any number } x$$

In this, when we take different numbers as x, we get many numbers as $x + 1$, $x + 2$ and $x^2 + 3x + 2$; and this equation shows the relation between these numbers.

We can consider all these algebraic expressions as polynomials. Then the above equation can be interpreted like this:

The product of the first degree polynomials $x + 1$ and $x + 2$ is the second degree polynomial $x^2 + 3x + 2$.

Like this, can you state the algebraic identity got from the first set of problems in terms of polynomial multiplication?

Let's look at the algebraic identities got from the different problems done so far:

- $(x + 2)(x + 8) = x^2 + 10x + 16$
- $(x + 3)(x + 7) = x^2 + 10x + 21$
- $(x + 1)(x + 2) = x^2 + 3x + 2$

In all these, what are the relations between the numbers in the first degree polynomials multiplied and the second degree polynomial got as the product?

Can you write a very general algebraic identity which includes all these products?

For any numbers x, a, b

$$(x + a)(x + b) = x^2 + (a + b)x + ab$$

How do we state this in terms of polynomials?

The product of the first degree polynomials $x + a$ and $x + b$ is the second degree polynomial $x^2 + (a + b)x + ab$

How do we write $(x + 5)(x + 3)$ using this?

$$(x + 5)(x + 3) = x^2 + 8x + 15$$

We have seen general identities on not only the product of sums, but also on the product of a sum and difference and product of differences in the lesson **Multiplication Identities** in the Class 9 textbook:

- $(x + y)(u - v) = xu - xv + yu - yv$
- $(x - y)(u - v) = xu - xv - yu + yv$

Using these, we can write the products of some other kinds of first degree polynomials:

- $(x + a)(x - b) = x^2 + (a - b)x - ab$
- $(x - a)(x - b) = x^2 - (a + b)x + ab$

For example,

$$(x + 4)(x - 1) = x^2 + 3x - 4$$

We can do this in another way. We can write

$$x - 1 = x + (-1)$$

so that

$$(x + 4)(x - 1) = (x + 4)(x + (-1))$$

Now in the identity

$$(x + a)(x + b) = x^2 + (a + b)x + ab$$

for the multiplication of sums, if we take a as 4 and b as -1, we get

$$\begin{aligned}(x + 4)(x + (-1)) &= x^2 + (4 + (-1))x + (4 \times (-1)) \\ &= x^2 + 3x - 4\end{aligned}$$

Thus we get as before

$$(x + 4)(x - 1) = x^2 + 3x - 4$$

Like this, the product $(x - 4)(x - 1)$ can be computed in two ways

In the identity

$$(x - a)\,(x - b) = x^2 - (a + b)x + ab$$

for the multiplication of differences, take a as 4 and b as 1 to get

$$(x - 4)(x - 1) = x^2 - 5x + 4$$

When we write the difference $x - 4$ as the sum $x + (-4)$ and difference $x - 1$ as the sum $x + (-1)$ we get,

$$(x - 4)(x - 1) = (x + (-4))(x + (-1))$$

Or in the identity

$$(x + a)(x + b) = x^2 + (a + b)x + ab$$

for the multiplication of sums, take a as -4 and b as -1 to get

$$\begin{aligned}(x - 4)(x - 1) &= (x + (-4))(x + (-1)) \\ &= x^2 + ((-4) + (-1))x + ((-4) \times (-1)) \\ &= x^2 - 5x + 4\end{aligned}$$

Find the following products

(i) $(x + 2)(x + 5)$ (ii) $(x + 2)(x - 5)$ (iii) $(x - 2)(x + 5)$ (iv) $(x - 2)(x - 5)$

Polynomial factors

We have done several problems on calculating the product of two first degree polynomials as a second degree polynomial. Now a question in reverse:

Can we write $x^2 + 5x + 6$ as the product of two first degree polynomials?

Here the product is given and we have to find the polynomials to be multiplied; thus we can rephrase the question like this:

To get $(x + a)(x + b) = x^2 + 5x + 6$, what numbers should we take as a and b?

We can write the product $(x + a)\ (x + b)$ as

$$(x + a)(x + b) = x^2 + (a + b)x + ab$$

So, the question can be written like this:

To get $x^2 + (a + b)x + ab = x^2 + 5x + 6$ what numbers should we take as a and b?

For this equation to hold, it is enough if we have $a + b = 5$ and $ab = 6$. So, we can change the question to this:

To get $a + b = 5$ and $ab = 6$, what numbers should we take as a and b?

In ordinary language, this means,

What two numbers have sum 5 and product 6?

A little thought would give the numbers as 2 and 3. So, the answer to the very first question is

$$x^2 + 5x + 6 = (x + 2)(x + 3)$$

Suppose we slightly change our original question like this:

Write $x^2 - 5x + 6$ as the product of two first degree polynomials.

As in the first problem, we can rewrite the question step by step:

(i) To get $(x + a)(x + b) = x^2 - 5x + 6$, what numbers should we take as a, b?

(ii) To get $x^2 + (a + b)x + ab = x^2 - 5x + 6$, what numbers should we take as a and b?

What can we say about $a + b$ and ab from this?

So, we can continue our thoughts like this:

(iii) To get $a + b = -5$ and $ab = 6$ what numbers should we take as a and b?

(iv) What two numbers have sum -5 and product 6?

If we take as before the numbers 2 and 3 giving the product 6, the sum would not be -5

How do we write 6 as a product of negative numbers?

$$6 = (-2) \times (-3)$$

(The section **Time and distance** of the lesson, **Negative Numbers** in the Class 9 textbook).

Moreover, we do have

$$(-2) + (-3) = -2 - 3 = -5$$

(The section **Position and number** of the lesson, **Negative Numbers** in the Class 9 textbook).

Thus the numbers in our last question above are -2 and -3.

And the answer to the first question is

$$\begin{aligned} x^2 - 5x + 6 &= (x + (-2))(x + (-3)) \\ &= (x - 2)(x - 3) \end{aligned}$$

Now what if we change the polynomial in the question like this?

Write $x^2 + 5x - 6$ as the product of two first degree polynomials.

How do we rewrite the question step by step?

(i) To get $(x + a)(x + b) = x^2 + 5x - 6$, what numbers should we take as a and b?

(ii) To get $x^2 + (a + b)x + ab = x^2 + 5x - 6$, what numbers should we take as a and b?

(iii) To get $a + b = 5$ and $ab = -6$, what numbers should we take as a and b?

(iv) What two numbers have sum 5 and product -6?

Now if we write $-6 = (-2) \times 3$ or $-6 = 2 \times (-3)$, we don't get the sum of the factors as -5.

So we must write 6 as the product of two other factors.

Using the fact that $6 = 1 \times 6$, we get these factorizations:

$$-6 = (-1) \times 6$$
$$-6 = 1 \times (-6)$$

If we take the first of these, we do get the sum of the factors as 5.

Thus the numbers asked for in question (iv) above are 6 and −1 and the answer to the question about polynomial factorization is

$$x^2 + 5x - 6 = (x + 6)(x - 1)$$

Now can't you find the answer to this question without much thought?

Write $x^2 - 5x - 6$ as the product of two first degree polynomials

$$x^2 - 5x - 6 = (x - 6)(x + 1)$$

Like this, write the second degree polynomials below as the product of two first degree polynomials:

(i) $x^2 + x - 6$ (ii) $x^2 - x - 6$ (iii) $x^2 + 7x + 6$ (iv) $x^2 - 7x + 6$

Unification

To calculate the first degree factors of both $x^2 + 5x + 6$ and $x^2 - 5x + 6$, we wrote the polynomial in the form

$$(x + a)(x + b)$$

and proceeded to compute a and b

In the first problem, we got $a = 2$ and $b = 3$, both positive numbers.

In the second problem, we got $a = -2$ and $b = -3$, both negative.

In the third problem, we got $a = 6$ and $b = -1$, a positive, b negative.

In the fourth problem, we got $a = -6$ and $b = 1$, a negative, b positive.

In short, to find the first degree factors of any second degree polynomial, we need only the single identity

$$(x + a)(x + b) = x^2 + (a + b)x + ab$$

We have this convenience because we use negative numbers also.

In all these, we found the numbers in the first degree polynomials simply by mental computations.

This may not be possible always. For example, see this problem:

Write $x^2 + 27x + 180$ as the product of two first degree polynomials.

To do this, we first write

$$(x + a)(x + b) = x^2 + 27x + 180$$

That is,

$$x^2 + (a + b)x + ab = x^2 + 27x + 180$$

From this we get

$$a + b = 27$$
$$ab = 180$$

If we go by the method used in the problems so far, we have to factorize 180 as the product of two numbers in different ways, and check which pair gives the sum 27; but this is not very easy. So, let's see if there is a different method.

Here we know the sum and product of two numbers. If it were the sum and difference, we could have found the numbers (the last part of the section **Two equations** of the lesson **Pairs of Equations** in the Class 9 textbook).

Can we calculate the difference of two numbers using their sum and product?

Haven't we seen in Class 8, how we can calculate the square of the difference of two numbers from the square of their sum and product? (The section **Square of difference** in the lesson **Square Identities**).

$$(a - b)^2 = (a + b)^2 - 4ab$$

Using this in our current problem,

$$\begin{aligned}(a - b)^2 &= 27^2 - (4 \times 180)\\ &= (400 + 280 + 49) - (72 \times 10)\\ &= 729 - 720\\ &= 9\end{aligned}$$

So what numbers can $a - b$ be?

$$a - b = 3 \text{ or } a - b = -3$$

Taking $a - b = 3$, we have

$$\begin{aligned}a + b &= 27\\ a - b &= 3\end{aligned}$$

From these we get

$$\begin{aligned}a &= \tfrac{1}{2}(27 + 3) = \tfrac{1}{2} \times 30 = 15\\ b &= \tfrac{1}{2}(27 - 3) = \tfrac{1}{2} \times 24 = 12\end{aligned}$$

If we take $a - b = -3$, we get instead

$$\begin{aligned}a + b &= 27\\ a - b &= -3\end{aligned}$$

and from these we get

$$\begin{aligned}a &= \tfrac{1}{2}(27 - 3) = \tfrac{1}{2} \times 24 = 12\\ b &= \tfrac{1}{2}(27 + 3) = \tfrac{1}{2} \times 30 = 15\end{aligned}$$

In either case, we get

$$x^2 + 27x + 180 = (x + 12)(x + 15)$$

In general, if we can write the polynomial $p(x)$ as $p(x) = q(x)r(x)$, the product of two polynomials $q(x)$ and $r(x)$, then the polynomials $q(x)$ and $r(x)$ are called *factors* of the polynomial $p(x)$ (In the case of numbers also, the meaning of factors is the same, isn't it? For example, since $10 = 2 \times 5$, we call 2 and 5 as factors of 10).

So, in all the problems above, what we have done is to factorize a second degree polynomial as the product of two first degree factors.

Write the polynomials below as the product of two first degree factors:

(i) $x^2 + 30x + 221$ (ii) $x^2 + 4x - 221$ (iii) $x^2 + x - 156$

Factors and solutions

Let's look at a problem we did in the section **Two answers** of the lesson **Second degree equations**

To get $x^2 - 4x + 3 = 0$, what number should we take as x?

How did we do this?

(i) $x^2 - 4x + 3 = 0$

(ii) $x^2 - 4x + 4 = 1$

(iii) $(x - 2)^2 = 1$

(iv) $x - 2 = 1$ or $x - 2 = -1$

(v) $x = 3$ or $x = 1$

We can do this in another way. For that, we first write the polynomial $x^2 - 4x + 3$ in the problem as the product of two first degree polynomials.

This we can do in head, right?

$$x^2 - 4x + 3 = (x - 1)(x - 3)$$

So we can rewrite our problem like this:

To get $(x-1)(x-3)=0$, what number should we take as x ?

For the product of two numbers to be zero, at least one of them should be zero, right?

So, to get $(x-1)(x-3)=0$, we should have either $x-1=0$ or $x-3=0$

In other words, either we must take $x=1$ or $x=3$

We can condense the steps of this procedure like this:

(i) $x^2-4x+3=0$

(ii) $x^2-4x+3=(x-1)(x-3)$

(iii) $(x-1)(x-3)=0$

(iv) $x-1=0$ or $x-3=0$

(v) $x=1$ or $x=3$

Let's slightly change the problem like this:

To get $x^2-5x+6=0$, what number should we take as x?

How do we write x^2-5x+6 as the product of two first degree polynomials?

$$x^2-5x+6=(x-2)(x-3)$$

Now how do we write the steps leading to the solution?

(i) $x^2-5x+6=0$

(ii) $x^2-5x+6=(x-2)(x-3)$

(iii) $(x-2)(x-3)=0$

(iv) $x-2=0$ or $x-3=0$

(v) $x=2$ or $x=3$

In the first problem, we found the solutions of the second degree equation $x^2-4x+3=0$ as 1 and 3; in the second problem, we found the solutions of the second degree equation $x^2-5x+6=0$ as 2 and 3

Thus we can solve any second degree equation by factorizing the polynomial, instead of completing the square as done earlier.

For example, see this problem:

One side of a rectangle is 3metres longer than the other side and its area is 270 square metres. What are the lengths of the sides?

If we denote the length of the shorter side as x metres, then the length of the longer side is $x + 3$ metres and its area is

$$x(x + 3) = x^2 + 3x \text{ square metres.}$$

So the problem can be stated using algebra like this:

To get $x^2 + 3x = 270$, what number should we take as x?

To do this using the earlier method of completing the square, we must add $\left(\frac{3}{2}\right)^2$ and proceed.

Let's try the factorization method instead.

To do this, we first rewrite the problem like this:

To get $x^2 + 3x - 270 = 0$, what number should we take as x?

Next we must factorize $x^2 + 3x - 270$. For this we write

$$x^2 + 3x - 270 = (x + a)(x + b) = x^2 + (a + b)x + ab$$

and from this get

$$a + b = 3$$
$$ab = -270$$

We now compute $a - b$ using these:

$$(a - b)^2 = (a + b)^2 - 4ab = 9 - (4 \times (-270)) = 9 - (-1080) = 9 + 1080 = 1089$$

To get $a - b$ we must calculate the square root of 1089. We have seen how square roots can be computed through factorization in class 7 (the section **Areas of squares** of the lesson **Squares And Right Triangles**).

Since the sum of the digits of 1089 is 18 which is a multiple of 9, the number itself is a multiple of 9 (the section **Digit sum** of the lesson **Number Relation** in the Class 6 textbook).

So, let's compute

$$1089 \div 9$$

We have done such computations in Class 5.

	9	
	1089–	
100	900	$9 \times 100 = 900$
	189–	
20	180	$9 \times 20 = 180$
	9–	
1	9	$9 \times 1 = 9$
$1089 \div 9 = 121$	0	$9 \times 121 = 1089$

Thus we get

$$1089 = 9 \times 121$$

This we can write like this:

$$1089 = 3^2 \times 11^2 = (3 \times 11)^2 = 33^2$$

(The section **Multiples and Powers** of the chapter **Repeated Multiplication** in the Class 7 textbook). What can we say from the equation $(a - b)^2 = 33^2$?

$$a - b = 33 \text{ or } a - b = -33$$

If we take $a - b = 33$, we get

$$a + b = 3$$
$$a - b = 33$$

From this we get

$$a = \frac{1}{2}(3 + 33) = \frac{1}{2} \times 36 = 18$$
$$b = \frac{1}{2}(3 - 33) = \frac{1}{2} \times (-30) = -15$$

Thus we have

$$x^2 + 3x - 270 = (x - 15)(x + 18)$$

So our problem can be now rewritten like this:

To get $(x - 15)(x + 18) = 0$, what number should we take as x?

As seen in earlier problems, the answer to this is $x = 15$ or $x = -18$

In our problem, x is the length of a side of a rectangle and so cannot be a negative number. So, the length of the shorter side of the rectangle is 15 metres.

The length of the longer side is $15 + 3 = 18$ metres.

Check what we get if in this problem if we proceed with $a - b = -33$ instead of $a - b = 33$.

Let's look at another problem:

One added to a number gives the square of the number. What is the number?

Taking the number as x, the problem can be written in algebra like this:

To get $x + 1 = x^2$, what number should we take as x?

To get $x + 1 = x^2$, we must have $x^2 - (x + 1) = 0$. In this,

$$x^2 - (x + 1) = x^2 - x - 1$$

(We have seen in class 7 that instead of subtracting a sum, we can subtract the addends one by one, in the section **One by one and altogether** of the lesson **Shorthand Math**).

So, we can rewrite our question like this:

To get $x^2 - x - 1 = 0$, what number should we take as x?

Next we factorize $x^2 - x - 1$ by writing

$$x^2 - x - 1 = (x + a)(x + b) = x^2 + (a + b)x + ab$$

and from this we get

$$a + b = -1$$
$$ab = -1$$

So we can calculate

$$(a - b)^2 = (a + b)^2 - 4ab = (-1)^2 - (4 \times (-1))$$
$$= 1 - (-4) = 1 + 4 = 5$$

and this gives

$$a - b = \sqrt{5} \text{ or } a - b = -\sqrt{5}$$

Taking $a - b = \sqrt{5}$ we get

$$a + b = -1$$
$$a - b = \sqrt{5}$$

Now we can compute a and b as

$$a = \frac{1}{2}(-1 + \sqrt{5}) = \frac{\sqrt{5} - 1}{2}$$
$$b = \frac{1}{2}(-1 - \sqrt{5}) = -\frac{\sqrt{5} + 1}{2}$$

Thus we get

$$x^2 - x - 1 = \left(x + \frac{\sqrt{5} - 1}{2}\right)\left(x - \frac{\sqrt{5} + 1}{2}\right)$$

Role reversal

We write a second degree polynomial $p(x)$ as the product of two first degree polynomials

$$p(x) = (x + a)(x + b)$$

and compute a and b.

In such problems, we first get $a + b$ and then $(a - b)^2$. If they are of the form

$$a + b = h$$
$$(a - b)^2 = k^2$$

We can take $a - b$ as either k or $-k$. If we take $a + b = h$ and $a - b = k$, we get

$$a = \frac{1}{2}(h + k)$$
$$b = \frac{1}{2}(h - k)$$

What if we take $a - b = -k$?

$$a = \frac{1}{2}(h - k)$$
$$b = \frac{1}{2}(h + k)$$

Thus the numbers a and b are interchanged; the factors however do not change.

Now the question becomes this:

$$\text{To get } \left(x+\frac{\sqrt{5}-1}{2}\right)\left(x-\frac{\sqrt{5}+1}{2}\right)=0 \text{ what should we take as } x?$$

The answer to this is

$$x=-\frac{\sqrt{5}-1}{2} \text{ or } x=\frac{\sqrt{5}+1}{2}$$

This means the speciality of the positive number $\frac{\sqrt{5}+1}{2}$ and the negative number $-\frac{\sqrt{5}-1}{2}$ is that 1 added to either of them gives its square.

(1) One side of a rectangle is 2 metres longer than the other and its area is 48 square metres. What are the lengths of its sides?

(2) One side of a right triangle is one centimetre less than twice the length of the side perpendicular to it; the hypotenuse is one centimetre more than twice this side. What are the lengths of the sides?

(3) The sum of the consecutive natural numbers from 1up to which number is 300?

(4) At what points does the graph of the polynomial $x^2 - 2x - 1$ cross the x-axis?

General solution

See this problem:

At what points does the graph of the polynomial $2x^2 + 3x - 2$ cross the x-axis?

We can rephrase the question like this:

To get $2x^2 + 3x - 2 = 0$, what number should we take as x?

This is a bit different from similar problems we have done earlier. In all the problems done so far we had only x^2 in the polynomial; here it is $2x^2$

So we first write $2x^2 + 3x - 2$ as

$$2x^2 + 3x - 2 = 2\left(x^2+\frac{3}{2}x-1\right)$$

From this we can see that if we take any number x which makes $x^2+\frac{3}{2}x-1=0$; also makes twice this polynomial $2x^2 + 3x - 2 = 0$; on the other hand, any number x which makes $2x^2 + 3x - 2 = 0$, also makes half this polynomial $x^2+\frac{3}{2}x-1=0$.

So we can again rewrite our problem like this:

To get $x^2 + \frac{3}{2}x - 1 = 0$, what number should we take as x?

Now we can find x as before using factorization. Writing

$$x^2 + \frac{3}{2}x - 1 = (x + a)(x + b) = x^2 + (a + b)x + ab$$

we get

$$a + b = \frac{3}{2}$$

$$ab = -1$$

and from this

$$(a - b)^2 = (a + b)^2 - 4ab = \frac{9}{4} - (4 \times (-1)) = \frac{9}{4} + 4 = \frac{25}{4}$$

So we have

$$a - b = \frac{5}{2} \text{ or } a - b = -\frac{5}{2}$$

Taking $a - b = \frac{5}{2}$, we get

$$a + b = \frac{3}{2}$$

$$a - b = \frac{5}{2}$$

and this gives

$$a = \frac{1}{2} \times \left(\frac{3}{2} + \frac{5}{2}\right) = \frac{1}{2} \times 4 = 2$$

$$b = \frac{1}{2} \times \left(\frac{3}{2} - \frac{5}{2}\right) = \frac{1}{2} \times (-1) = -\frac{1}{2}$$

Thus we find

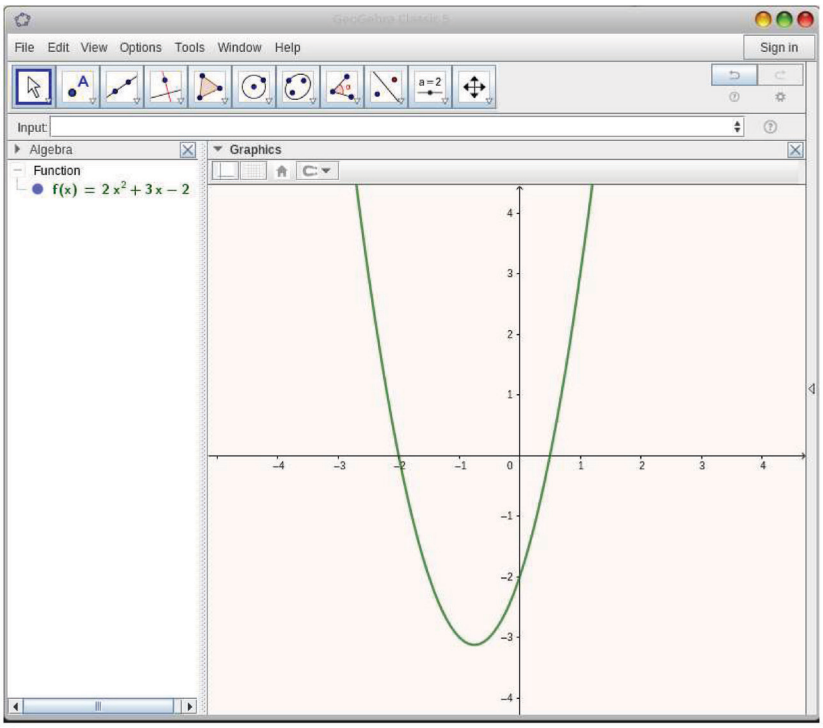
$$x^2 + \frac{3}{2}x - 1 = (x + 2)\left(x - \frac{1}{2}\right)$$

So, the problem becomes this:

To get $(x + 2)\left(x - \frac{1}{2}\right) = 0$, what number should we take as x?

The answer to this is $x = -2$ or $x = \frac{1}{2}$.

Thus the points where the graph of the polynomial $2x^2 + 3x - 2$ crosses the x-axis are $(-2, 0)$ and $\left(\frac{1}{2}, 0\right)$.



Now let's look at this method in general terms:

Any second degree polynomial can be written in the form $ax^2 + bx + c$. So any second degree equation can be written in the form

$$ax^2 + bx + c = 0$$

The method of computing the solutions by factorization can be split into stages like this:

(1) Change the equation to $x^2 + \frac{b}{a}x + \frac{c}{a} = 0$

(2) Write the polynomial in this as the product of first degree factors:

(i) Writing the factors as $x + p$ and $x + q$, we write

$$x^2 + \frac{b}{a}x + \frac{c}{a} = (x + p)(x + q) = x^2 + (p + q)x + pq$$

(ii) From this we get $p + q = \frac{b}{a}$ and $pq = \frac{c}{a}$.

(iii) Use the identity $(p - q)^2 = (p + q)^2 - 4pq$ to compute $p - q$.

(iv) Use $p + q$ found earlier and one of the values of $p - q$ to calculate p and q.

(3) Then the equation to be solved can be written as $(x+p)(x+q)=0$.

(4) The solutions to this are $x=-p$ or $x=-q$.

Now let's see in detail how $p-q$ and then p and q are calculated:

$$(p-q)^2 = \left(\frac{b}{a}\right)^2 - 4\left(\frac{c}{a}\right)$$
$$= \frac{b^2}{a^2} - \frac{4c}{a} = \frac{b^2}{a^2} - \frac{4ac}{a^2} = \frac{b^2-4ac}{a^2}$$
$$p-q = \frac{\sqrt{b^2-4ac}}{a} \text{ or } p-q = -\frac{\sqrt{b^2-4ac}}{a}$$

We can calculate p and q by taking either of these as $p-q$.

$$p+q = \frac{b}{a} \qquad p-q = \frac{\sqrt{b^2-4ac}}{a}$$
$$p = \frac{1}{2}\left(\frac{b}{a} + \frac{\sqrt{b^2-4ac}}{a}\right) = \frac{1}{2}\times\frac{b+\sqrt{b^2-4ac}}{a} = \frac{b+\sqrt{b^2-4ac}}{2a}$$
$$q = \frac{1}{2}\left(\frac{b}{a} - \frac{\sqrt{b^2-4ac}}{a}\right) = \frac{1}{2}\times\frac{b-\sqrt{b^2-4ac}}{a} = \frac{b-\sqrt{b^2-4ac}}{2a}$$

Thus we get the solutions of the equation as

$$x = -\frac{b+\sqrt{b^2-4ac}}{2a} \text{ or } x = -\frac{b-\sqrt{b^2-4ac}}{2a}$$

This we can write as

$$x = \frac{-b-\sqrt{b^2-4ac}}{2a} \text{ or } x = \frac{-b+\sqrt{b^2-4ac}}{2a}$$

We use the symbol ± to denote adding or subtracting a number. So, we can write the solutions like this:

The solutions of the equation

$$ax^2 + bx + c = 0$$

are given by

$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

As an illustration let's have another look at an earlier problem:

At what points does the graph of the polynomial $2x^2 + 3x - 2$ cross the x-axis?

Here the equation to be solved is

$$2x^2 + 3x - 2 = 0$$

So we take in the general form of the solutions stated above

$$a = 2 \qquad b = 3 \qquad c = -2$$

to get the solutions as

$$x = \frac{-3 \pm \sqrt{3^2 - (4 \times 2 \times (-2))}}{2 \times 2}$$
$$= \frac{-3 \pm \sqrt{9+16}}{4}$$
$$= \frac{-3 \pm \sqrt{25}}{4}$$
$$= \frac{-3 \pm 5}{4}$$

That is either

$$x = \frac{-3+5}{4} = \frac{2}{4} = \frac{1}{2}$$

or

$$x = \frac{-3-5}{4} = \frac{-8}{4} = -2$$

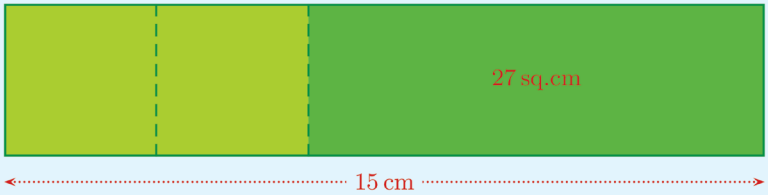
So the points where the graph crosses x - axis are $\left(\frac{1}{2}, 0\right), (-2, 0)$.

(1) Find the points at which the graphs of the polynomials given below cross the x-axis:

(i) $2x^2 - 7x - 1$ (ii) $2x^2 + 7x - 1$ (iii) $9x^2 + 12x + 4$

(2) The perimeter of a rectangle is 42 metres and its diagonal is 15 metres. What are the lengths of its sides?

(3) From a rectangular sheet of paper, two squares are cut off as shown below:



The area of remaining part is 27 square centimetres. What is the length of the shorter side of the rectangle?

(4) How many terms of the arithmetic sequence, 1, 5, 9, ... starting from the first, are to be added to get 91?

(5) A rectangle is to be made by bending a 28 centimetres long rod

(i) Can a rectangle of diagonal 8 centimetres be made?

(ii) How about a rectangle of diagonal 10 centimetres?

(iii) And a rectangle of diagonal 14 centimetres?

Calculate the lengths of the sides of those rectangles which can be made as above.